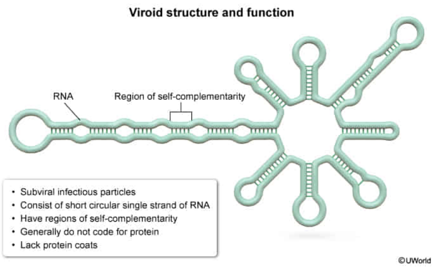
Scientists identify a new type of infectious agent that causes disease in humans. They hypothesize that this agent is a viroid. Which of the following observations best supports this hypothesis?

- A. The agent is surrounded by a protein coat that promotes entry into human cells. [31%]
- B. The agent lacks genetic material but can still replicate inside human cells. [17%]
- ✔ C. The agent is found to be composed entirely of a single circular RNA molecule. [42%]
- D. When exposed to the agent, endogenous human proteins become misfolded. [8%]

Correct
41% answered correctly

Explanation:



Viroids are subviral infectious particles that each consist of a short circular single strand of RNA. Due to this structure, viroids have regions where the RNA binds with itself (ie, self-complementarity), creating some double-stranded areas of the circular genome. The RNA composing viroids does not generally code for protein, and, unlike viruses, viroids lack protein coats (ie, capsids).

When infecting cells, viroids can bind host cellular RNA sequences, resulting in gene silencing that prevents synthesis of necessary proteins. Most known viroids infect plants; however, the pathogen causing hepatitis D is an example of a viroid capable of infecting humans. Viroid replication within a host cell is not well understood. It is thought to be similar to replication displayed by RNA viruses in some cases, while in others the viroid appears to cause the host cell's RNA polymerase to read RNA templates.

In this question, scientists identify a new type of infectious agent causing disease in humans, which is hypothesized to be a viroid. If the agent is found to be **composed entirely of a single circular RNA molecule**, this would provide support for the hypothesis because viroids consist of RNA alone.

(Choice A) Viruses are infectious agents consisting of genetic material (either DNA or RNA) surrounded by a protein coat. Viruses require the presence of a host cell in order to replicate. If the agent the scientists have identified is surrounded by a protein coat that aids its entry into human cells, this is consistent with a virus, *not* a viroid.

(Choices B and D) Prions are misfolded proteins that act as infectious agents by causing changes to the secondary structure of endogenous host proteins, which consequently become misfolded. These misfolded proteins aggregate and can cause disease. If the identified agent lacks genetic material but can still replicate in human cells and causes endogenous human proteins to become misfolded, this is consistent with a prion, *not* a viroid.

Educational objective:
Viroids are subviral infectious particles consisting of a short circular single strand of RNA. Unlike viruses, viroids are not surrounded by a protein coat. When infecting cells, viroids can bind host cellular RNA sequences, resulting in gene silencing that prevents synthesis of necessary proteins.

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